

Cluster-Randomised Trials

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Introduction

Cluster-randomised trials (CRTs) randomise groups – clinics, schools, villages – rather than individuals. Used when an intervention must be delivered at cluster level (implementation, educational campaign) or when contamination between individuals would bias a standard RCT. Clustering inflates variance and must be accounted for in sample-size and analysis.

Prerequisites

Intraclass correlation (ICC); mixed-effects models.

Theory

Design effect: $DE = 1 + (m - 1)\rho$, where m is average cluster size and ρ is the ICC. Effective sample size = actual N / DE . Sample-size calculations inflate by DE relative to individually-randomised trials.

Analysis accounts for clustering via mixed-effects models (random cluster intercept) or GEE with cluster-robust SE.

Assumptions

Clusters are exchangeable; intervention is applied uniformly within cluster; ICC estimate from pilot / literature is approximately correct.

R Implementation

```
library(lme4); library(lmerTest)

set.seed(2026)
# 20 clusters, avg 15 patients per cluster
n_clusters <- 20
m_per <- 15
cluster <- factor(rep(1:n_clusters, each = m_per))
arm <- factor(rep(c("ctrl", "trt"), each = (n_clusters/2) * m_per))
clust_re <- rep(rnorm(n_clusters, 0, 0.8), each = m_per)

y <- clust_re + ifelse(arm == "trt", 0.5, 0) +
  rnorm(n_clusters * m_per, 0, 1)

df <- data.frame(cluster, arm, y)
fit <- lmer(y ~ arm + (1 | cluster), data = df)
summary(fit)$coefficients

# Empirical ICC
vc <- as.data.frame(VarCorr(fit))
```

```
icc <- vc$vcov[1] / sum(vcov$vcov)
cat("Estimated ICC:", round(icc, 3), "\n")
```

Output & Results

Cluster-random-effect-adjusted treatment effect (~ 0.5) with SE accounting for clustering; ICC estimate ~ 0.4 .

Interpretation

“Cluster-randomised analysis estimated a 0.49 SD improvement (95 % CI 0.21-0.77, $p = 0.001$), accounting for the intra-cluster correlation of 0.42 via a random-cluster intercept.”

Practical Tips

- Even a small ICC (0.01) inflates required sample size substantially; budget accordingly.
- Pilot data or literature usually provides ICC; report both the planning value and the observed value.
- For few clusters (< 30), mixed-effects SE underestimates; use Kenward-Roger or Satterthwaite DF.
- Report per CONSORT extension for cluster trials; include number of clusters, cluster sizes, ICC.
- Stratified or matched-pair cluster designs improve balance when cluster count is small.

Reporting

A defensible cluster-trial report names the unit of randomisation, the unit of analysis, the planning ICC, and the achieved ICC, and explains how the analytical model handles their potential mismatch. Where the number of clusters is below thirty, state which small-sample correction was used for standard errors and degrees of freedom, since naive likelihood-based intervals are anti-conservative in this regime. If clusters varied substantially in size, mention whether weights were applied and how missing clusters or partial cluster dropout were handled, because differential cluster attrition can bias the estimated treatment effect even when individual-level missingness is modest.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale